Negative-index Lagrangian Fredholm pairs exist in weak symplectic Hilbert space

Literature-status note for arXiv:1301.7248

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Status: literature already answered (affirmative construction). This is an exact later-literature resolution, not a new run proof.

Original open problem

In Section 2.3.1, PDF page 14, Booß–Bavnbek and Zhu note that a Fredholm pair of Lagrangian subspaces can never have positive index. They then ask whether, unlike the strong case, a weak symplectic Hilbert space can have such a pair with negative index, and label this an open problem [1].

Decisive later answer

The same authors answer the question in Example 1.2.11, PDF pages 29–30, of their later paper [2]. Given $n \in \mathbb{N}$, they construct a complex Hilbert space

$$X = X_1 \oplus X_2 \oplus X_3, \qquad \dim X_1 = n, \qquad X_2 \simeq X_3,$$

and a bounded skew-self-adjoint injective but nonsurjective operator J such that $\omega(x, y) = \langle Jx, y\rangle$ is weak symplectic. Identifying X_2 and X_3, they define

$$\lambda_\pm = \{(\alpha, \pm\alpha) : \alpha \in X_2\}.$$

The example proves that $\lambda_\pm$ are Lagrangian in X, have trivial intersection, and have sum of codimension n. Hence

$$\operatorname{index}(\lambda_+, \lambda_-) = -n.$$

Thus negative indices not only exist: every negative integer occurs.

The supporting authors explicitly present Example 1.2.11 as resolving the phenomenon left open in their earlier paper.

Search evidence and scope

Exact searches for “Fredholm pairs of Lagrangian subspaces”, “negative index”, and “weak symplectic Hilbert space” in the local arXiv corpus and bounded web search found arXiv:1406.0569. Its introduction and Example 1.2.11 explicitly state the answer. No originality claim is made here.

This resolves only the negative-index existence problem. The neighboring questions in the original paper concerning projection symmetries, contractibility of the weak Lagrangian Grassmannian, and Bott periodicity are separate.

References

[1] B. Booß–Bavnbek and C. Zhu, *The Maslov index in weak symplectic functional analysis*, Ann. Global Anal. Geom. 44 (2013), 283–318; arXiv:1301.7248, Section 2.3.1.

[2] B. Booß–Bavnbek and C. Zhu, *The Maslov index in symplectic Banach spaces*, arXiv:1406.0569, Example 1.2.11.